

Exam-1 Computer Networking

1. What is a computer network?

A computer network is a **collection of two or more devices** (computer, laptop, server, printer, etc.) which are **connected together** to **share data, resources, and information** using wired or wireless media.

Example:

Sharing internet, files, printer in office or home network.

2. What are LAN, MAN, and WAN?

LAN (Local Area Network):

LAN is used for **small area** like home, office, school, building.
Speed is high and cost is low.

MAN (Metropolitan Area Network):

MAN covers **city level network**.

Example: Network connecting multiple branches in same city.

WAN (Wide Area Network):

WAN covers **large geographical area** like country to country.
Internet is the biggest example of WAN.

3. Difference between Hub, Switch, and Router?

Hub:

- Works on Physical layer
- Sends data to **all ports**
- Slow and more collision

Switch:

- Works on Data Link layer
- Sends data to **specific device**
- Uses MAC address

- Faster than hub

Router:

- Works on Network layer
 - Connects **different networks**
 - Uses IP address
 - Used to access internet
-

4. What is bandwidth and latency?

Bandwidth:

Bandwidth is the **maximum data transfer capacity** of a network.
Higher bandwidth means more data can pass.

Latency:

Latency is the **time delay** taken by data to travel from source to destination.
Lower latency means faster response.

5. What is a MAC address? Is it changeable?

MAC address (Media Access Control) is a **unique physical address** of NIC (Network Interface Card).
It is **48-bit hexadecimal** address.

Example: 00:1A:2B:3C:4D:5E

Normally MAC address is **not changeable**, but using software it can be **spoofed (temporarily changed)**.

6. What is the OSI model? Name all 7 layers.

OSI model (Open System Interconnection) explains **how data travels** from sender to receiver.

7 Layers (Top to Bottom):

1. Application
2. Presentation

3. Session
4. Transport
5. Network
6. Data Link
7. Physical

Mnemonic: **All People Seem To Need Data Processing**

7. What is an IP address?

IP address (Internet Protocol address) is a **logical address** assigned to a device to **identify and communicate** on a network.

Example: 192.168.1.10

8. Difference between IPv4 and IPv6?

IPv4	IPv6
32-bit address	128-bit address
Numeric format	Alphanumeric
Limited IPs	Huge IP addresses
Example: 192.168.1.1	Example: 2001:db8::1

9. What are public and private IP addresses?

Public IP:

Used on internet, globally unique, provided by ISP.

Private IP:

Used inside local network, not routable on internet.

10. What are the private IP address ranges?

- Class A: 10.0.0.0 – 10.255.255.255
 - Class B: 172.16.0.0 – 172.31.255.255
 - Class C: 192.168.0.0 – 192.168.255.255
-

11. What is APIPA?

APIPA (Automatic Private IP Addressing)

If DHCP server is not reachable, system assigns IP automatically.

Range: 169.254.0.0 – 169.254.255.255

12. What is static IP and dynamic IP?

Static IP:

Manually assigned, does not change.

Dynamic IP:

Automatically assigned by DHCP, may change.

13. What is a network ID and host ID?

Network ID: Identifies the network

Host ID: Identifies the device inside the network

Example:

IP: 192.168.1.10/24

Network ID: 192.168.1.0

Host ID: 10

14. What is broadcast IP address?

Broadcast IP is used to **send data to all devices** in the network.

Example:

Network: 192.168.1.0/24

Broadcast: 192.168.1.255

15. Can two devices have the same IP address? Why/why not?

No, two devices **cannot have same IP** in same network.
It causes **IP conflict**, network communication fails.

16. Default subnet mask for:

- Class A: 255.0.0.0
 - Class B: 255.255.0.0
 - Class C: 255.255.255.0
-

17. Why do we need a subnet mask?

Subnet mask is used to:

- Identify network and host portion
 - Divide network into subnets
 - Reduce broadcast traffic
-

18. Difference between FLSM and VLSM?**FLSM (Fixed Length Subnet Mask):**

- All subnets same size
- Wastes IP addresses

VLSM (Variable Length Subnet Mask):

- Different subnet sizes
 - Efficient IP usage
-

19. Four steps of DHCP process (DORA)

1. Discover
 2. Offer
 3. Request
 4. Acknowledge
-

20. Can DHCP assign IP to devices in different networks?

Yes, using **DHCP Relay (IP Helper Address)** DHCP can assign IP to different networks.

21. Subnetting for 4 departments with 50 users each

50 users need **64 IPs** → /26 subnet

Example network: 192.168.1.0/24

- Dept 1: 192.168.1.0/26
 - Dept 2: 192.168.1.64/26
 - Dept 3: 192.168.1.128/26
 - Dept 4: 192.168.1.192/26
-

22. Command to assign IP address to router interface

ip address 192.168.1.1 255.255.255.0

23. Purpose of no shutdown command

By default router interface is **administratively down**.
no shutdown command **enables the interface**.

24. Why router interface stays administratively down?

Because interface is **not enabled manually** using no shutdown.

25. Different router modes in Cisco IOS

- User EXEC mode (>)
- Privileged EXEC mode (#)
- Global Configuration mode
- Interface Configuration mode

26. Difference between enable password and enable secret?

- Enable password: Stored in plain text
- Enable secret: Encrypted (more secure)

27. Command to save running configuration

do write

OR

copy running-config startup-config

28. Configure IP on GigabitEthernet 0/0

enable

configure terminal

interface gigabitEthernet 0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

exit